

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

- A. It activates TLR5 on the surface of macrophages.
- B. It primarily induces the Th1-cell response.
- C. Peptidoglycan is its major antigen that induces an immune response.
- D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

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The patient is a 34-year-old poultry worker with a sore throat, non-productive cough, fever, oropharyngeal erythema, moist rales, and patchy reticular opacities on chest X-ray, consistent with community-acquired pneumonia (CAP). The clinical presentation and imaging findings are typical of a bacterial pneumonia, and the patient’s exposure to poultry suggests a possible exposure to bacteria common in such environments.

Among the common pathogens causing CAP in adults, especially in individuals with occupational exposure, *Streptococcus pneumoniae* is a frequent cause. However, given the exposure to poultry, *Haemophilus influenzae* or *Moraxella catarrhalis* are also possible. But the key clue is the presence of "patchy reticular opacities in the perihilar regions," which is characteristic of *Streptococcus pneumoniae* or other encapsulated bacteria.

Now, the question asks about the immune response elicited by the pathogen.

Let’s evaluate the options:

A. It activates TLR5 on the surface of macrophages.

TLR5 recognizes bacterial flagellin. This is relevant to flagellated bacteria (e.g., *Salmonella*, *E. coli*), not typically associated with common CAP pathogens like *S. pneumoniae* or *H. influenzae*. Not the best answer.

B. It primarily induces the Th1-cell response.

This is plausible. Bacterial pneumonia, especially from encapsulated bacteria like *S. pneumoniae*, triggers a strong innate immune response involving macrophages and dendritic cells, leading to the activation of T-helper1 (Th1) cells. Th1 responses are critical in fighting intracellular bacteria and promoting the production of cytokines like IFN- γ and IL-12, which activate macrophages. This is a hallmark of bacterial pneumonia.

C. Peptidoglycan is its major antigen that induces an immune response.

Peptidoglycan is a component of bacterial cell walls and is recognized by TLR2 and NOD1. While it is indeed a component of bacterial pathogens, it is not the best description of the immune response in the context of the immune response type (e.g., Th1). Also, this is a structural component, not a description of the immune response type.

D. This pathogen evades the immune response by encapsulation.

Encapsulation is a feature of *Streptococcus pneumoniae* and *Haemophilus influenzae*, and it helps evade phagocytosis by masking the bacterial surface. However, the question specifically asks for the immune response *elicited*, not immune evasion. So this option describes evasion, not the immune response.

Therefore, the best description of the *immune response* elicited by the pathogen is the activation of a Th1 response, which is typical in bacterial pneumonia.

< answer > B. It primarily induces the Th1-cell response. </answer >

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